Journal of
Materials Chemistry A

Cite this: *J. Mater. Chem. A*, 2013, **1**, 1108

Solid polymer electrolytes which contain tricoordinate boron for enhanced conductivity and transference numbers

Kayla L. Mathews,[a] Angela M. Budgin,[a] Srinivas Beeram,[a] Anisha T. Joenathan,[a] Barry D. Stein,[b] Ulrike Werner-Zwanziger,[c] Maren Pink,[a] Lane A. Baker,[a] Waleed E. Mahmoud,[d] John P. Carini*[e] and Lyudmila M. Bronstein*[ad]

Received 9th October 2012
Accepted 16th November 2012

DOI: 10.1039/c2ta00628f

www.rsc.org/MaterialsA

Here we report syntheses and study of composite solid polymer electrolytes (SPEs) based on a poly(ethylene glycol)-in-Li triflate material that contains an organic-inorganic composite (OIC) in which boron species are incorporated into a silica network. The structure and properties of the SPEs synthesized were characterized by scanning transmission electron microscopy (STEM), ^{29}Si, ^{11}B and ^{13}C solid state NMR, differential scanning calorimetry, and impedance spectroscopy. STEM allowed assessment of OIC particles in their native environment without removal of an organic component. The Lewis acid tricoordinate boron sites formed in OIC are proposed to have a stronger interaction with triflate anions than silica sites, which results in enhanced lithium ion conductivity and Li transference numbers at optimal boron concentrations. The optimum triethyl borate (TEB) concentration also leads to formation of smaller (higher surface area) OIC particles, which expose more boron sites to triflate anions. The SPE sample prepared with 10 mol% TEB exhibited a conductivity of 4.3×10^{-5} S cm^{-1} and a Li transference number of 0.89, which represents nearly single-ion conductor behaviour for the salt-in-polymer–borosilicate composite.

1 Introduction

Over the past decade, solid polymer electrolytes (SPEs) have received considerable attention for application in diverse technologies, such as rechargeable batteries, fuel cells, supercapacitors, *etc.*[1–3] Much of this attention is a result of efforts to miniaturize devices and because of the exceptional stability of SPEs, in comparison to liquid electrolytes.

Poly(ethylene oxide) (PEO) is the most commonly used polymer, however because of crystallinity, the conductivity of PEO based electrolytes is low at ambient temperatures.[4,5] Various methods have been used to suppress crystallinity and thereby increase conductivity as a result of higher polymer chain mobility and the resultant faster cation diffusion. Crystallization has been suppressed in PEO block copolymers,[6,7] through crosslinks,[8,9] in polymer grafts,[10,11] polymer blends,[12–18] and due to the introduction of inorganic fillers.[4,19] In addition to suppression of crystallization, fillers can have other effects on electrochemical performance. For example, conductivity enhancements observed for Al_2O_3 nanoparticles in a PEO film were believed to be a result of both decrease in the crystallinity of the film and Lewis acid–base interactions from Lewis acid sites present on the nanoparticle surface. A similar behavior was also observed when SiO_2,[20] α-Al_2O_3,[21] $AlBr_3$ [22] and ZnO[23] nanoparticles were incorporated. Lewis acidity promotes dissociation of lithium salts through interactions with counter ions, thus facilitating transport of Li ions.[24]

Apart from conductivity, another important factor to consider is the lithium transference number which defines the contribution to conductivity from only lithium cations. Normally, lithium ion transport does not occur solely as free cations but in aggregates of ions of various sizes.[25] This aggregation results in low Li ion transference numbers, which is particularly disadvantageous in battery applications because more energy, time, and electrochemical potential would be required to recharge the battery.[26] The ideal transference number is unity, which is normally achieved only for single ion conductors. This can be accomplished by mixing ion-conducting polymers and polyelectrolytes, but mixing often results in compatibility issues.[26–30] Recent examples of solid polymer single ion conductors were reported for lithium poly(4-styrenesulfonyl-(trifluoromethylsulfonyl)imide)[31] and lithium oxalate polyacrylic acid borate[32] however in both cases the conductivities at

[a]Department of Chemistry, Indiana University, Bloomington, IN 47405, USA. E-mail: lybronst@indiana.edu; Tel: +1 812 855 3727
[b]Department of Biology, Indiana University, Bloomington, IN 4740, USA
[c]Department of Chemistry and Institute for Research in Materials, Dalhousie University, Halifax, NS B3H 4J3, Canada
[d]Department of Physics, Faculty of Science, King Abdulaziz University, Jeddah, Saudi Arabia
[e]Department of Physics, Indiana University, Bloomington, IN 47405, USA. E-mail: jcarini@indiana.edu

room temperature were much below 10^{-5} S cm^{-1}. Single ion conductance is also obtained for intercalates with clay particles but this results in low conductivity and anisotropy.[33–36] In some cases, single ion conductors do not allow good mechanical performance.[37]

The goal of this work was to increase lithium transference in composite SPEs through incorporation of inorganic particles possessing tricoordinate boron sites, which can act as Lewis acids. In previous papers we reported enhancement of conductivity for SPEs that contained aluminosilicate or silicate organic–inorganic component (OIC) particles formed *in situ* within PEO compared to other composite SPEs.[38–41] *In situ* formation resulted in fresh interfaces that interacted better with anions and led to a lithium ion transference number of 0.8 due to strong interactions of triflate anions with silicate particles.[39]

As boron species are typically stronger Lewis acids than alumina or silica, SPEs that contain boron have attracted a great deal of attention. A number of papers have been published where boron containing compounds were incorporated as low molecular weight additives[42,43] or boron species were imbedded in a polymer.[44] An example of the former case are plasticizers made of PEG-borate and PEG-aluminate esters which have shown good conductivity enhancements due to Lewis acidity of boron and aluminum,[42] but transference observed was small due to mobility of the plasticizers that could sequester anions. On the other hand, when poly(lithium carboxylate) (polymeric anion) was used, the addition of $BF_3 \cdot OEt_2$ stimulated dissociation of Li salt, while the transport of anions was suppressed that resulted in transference numbers 0.41–0.70.[44]

Incorporation of anion trapping boroxine rings into a polymer host resulted in a significant increase of Li transference numbers although conductivity was mediocre.[45] Boron moieties inserted in the polymer chain *via* interaction of phenylboronic acid with PEO "spacer" units of different length led to higher conductivities and transference numbers.[46] Improvement of Li transference numbers up to 0.82 was obtained by polymerization of mesitylborane with monomers with an oligo(ethylene oxide) unit,[47,48] however, high transference numbers were obtained only at low conductivities. Hybrid polymers with a tetraarylpentaborate unit demonstrated comparatively high conductivity for single ion conductors,[49] but real transference was not measured.

Here we report on a novel and facile approach for the introduction of boron species into composite SPEs. In this approach a strong Lewis acid, triethyl borate (TEB), is reacted with OIC nanoparticles (NPs) formed *in situ* within poly(ethylene glycol) (PEG) of 600 Da that contains Li triflate. We demonstrate that both the concentration of boron moieties and the manner in which TEB is added influence the electrochemical behavior of SPEs. The structures and properties of SPEs synthesized were characterized with scanning transmission electron microscopy (STEM), ^{29}Si, ^{11}B and ^{13}C solid state NMR, Fourier transform infrared spectroscopy (FTIR), differential scanning calorimetry (DSC), and AC impedance spectroscopy. STEM allowed characterization of the composite SPEs for the first time in their native condition (without removal of organic component) and estimation of OIC NP sizes. We demonstrate that an optimal amount of TEB allows formation of the smallest NPs allowing high conductivity and Li transference (0.89), which is unique for non-single-ion conductors.

2 Experimental

2.1 Materials

Poly (ethylene glycol) (PEG) with molecular weight 600 Da was purchased from Aldrich and used as received. Lithium triflate ($LiCF_3SO_3$, LiTf), (96%), aluminum-tri-*sec*-butoxide (AB) (97%), 3-glycidyloxypropyltrimethoxysilane (GLYMO) (≥98%), tetramethyl orthosilicate (TMOS) (≥99%), triethyl borate (TEB) (≥99%), tetrahydrofuran (THF) (≥99.9%), were purchased from Aldrich. Chloroform (99.8%) was purchased from MACRON Chemicals and also used without purification. Water was purified using a Milli-Q millipore purification system.

2.2 Synthesis of SPEs

2.1.1 Synthesis of SPEs with silica OIC. SPEs with silica OIC were synthesized in accordance with procedures described elsewhere.[38,39] In a typical experiment, 0.8 g (18 mmol) of PEG dissolved in 5 mL of chloroform was mixed with 0.2 g (1.28 mmol) of LiTf dissolved in 5 mL of THF. This amounts to a molar ratio Li : O (ether) = 1 : 14. In our earlier papers this was found to be an optimal ratio for composite SPEs and it compares well with other SPEs based on PEG.[37] The resultant solution was stirred for 1 hour. The OIC was prepared by sol–gel reaction of GLYMO and TMOS at a molar ratio of 4 : 1, where a vial which contained a stir bar was charged with 2.66 g (11.25 mmol) GLYMO, 0.426 g (2.81 mmol) TMOS, and 0.04 g (0.16 mmol) AB which served as a catalyst of epoxy group polymerization. Hydrolysis of the mixture was initiated by addition of 15% v/w (0.12 mL) HCl (0.01 N solution) needed for complete hydrolysis. After stirring the solution at room temperature for 30 min, the reaction mixture was charged with the rest of the 0.01 N HCl solution (0.64 mL) and stirred for 40 minutes at room temperature. The temperature of the reaction mixture was then increased to 50 °C and stirring was continued for another 15 min. A calculated amount of this precondensed silica solution was added to the PEG and Li salt solution. The amount was determined from the ratio of OIC to PEG + LiTf, and was based on 55% of OIC by weight in the SPE. The solution was then heated on a Teflon dish at 70 °C and left overnight for solvent evaporation and OIC condensation. The solid film was subjected to complete condensation by heating at 130 °C in a vacuum oven for 1 hour. Films were then removed from the dish, and placed in a dessicator before further use. SPEs based on GLYMO, TMOS, and AB are abbreviated as GTA (see Table 1).

2.1.2 Incorporation of boron species in OIC. Boron species were incorporated in OIC by four different methods (Table 1). In method A, the OIC mixture was placed in an ice bath after heating at 50 °C, TEB was added and the solution left to stir for 30 minutes; the rest of the procedure continued as described above. In method B, TEB was added after AB in the OIC and the procedure continued as above. In method C, TEB was added after AB in the OIC, the mixture was cooled to 0 °C in

Table 1 DSC data and OIC NP sizes for the SPEs with different amounts of TEB prepared by different methods

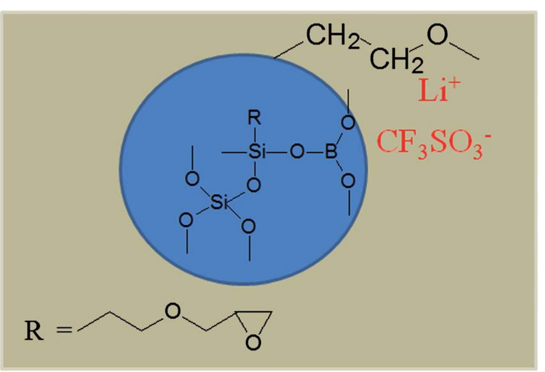
Sample	% TEB	Method	T_g (°C)	NP size (nm)/ std. dev. (%)	Conductivity (S cm^{-1})
GTA-PEG-1	0	n/a	−43	214/53	7.12×10^{-6}
GTA-PEG-2	5	C	−47	203/35	8.53×10^{-6}
GTA-PEG-3	10	A	−51	200/27	6.19×10^{-6}
GTA-PEG-4	10	B	−57	204/57	3.46×10^{-6}
GTA-PEG-5	10	C	−60	36/58	4.34×10^{-5}
GTA-PEG-6	10	D	−52	134/23	1.25×10^{-5}
GTA-PEG-7	15	C	−57	90/40	1.75×10^{-5}
GTA-PEG-8	20	C	−55	147/52	1.64×10^{-5}
GTA-PEG-9	25	C	−50	156/34	1.52×10^{-5}
GTA-PEG-10	30	C	−50	167/38	1.55×10^{-5}

an ice bath, and 0.12 mL of HCl (0.01 N) was added and the solution stirred for 15 minutes at 0 °C and 15 minutes at room temperature. The reaction mixture was then charged with the residual 0.01 N HCl solution (0.64 mL) and stirred for 40 minutes and the procedure continued as above. In method D, after the OIC mixture was stirred for 30 minutes, TEB was added and the solution stirred for 15 minutes, then 0.64 mL of HCl (0.01 N) was added and the solution was stirred for 40 minutes, and then the procedure continued as above. TEB was incorporated in amounts of 5, 10, 15, 20, 25, and 30 mol% of OIC. The GLYMO amount was decreased respectively, while the amounts of TMOS and AB stayed unchanged. For example, in the 5 mol% TEB sample, 0.103 g (0.705 mmol) of TEB was added along with 2.504 g (10.595 mmol) of GLYMO, while in the 10 mol% TEB sample, 0.206 g (1.41 mmol) of TEB was added along with 2.337 g (9.89 mmol) of GLYMO, *etc.* Scheme 1 shows a schematic representation of SPE based on PEG, Li triflate and TEB-modified OIC particles.

Please note that for each sample, at least three experiments have been carried out and the results agreed within 10% error. The results presented in Table 1 are averaged over three measurements for each sample.

2.3 Characterization

The ground SPE samples were embedded in epoxy resin and subsequently microtomed at ambient temperature. STEM



Scheme 1 Schematic representation of the SPE (gray rectangle) based on PEG, Li triflate and TEB-modified OIC particles (blue circle).

images of the resultant thin sections (*ca.* 50 nm thick) were obtained at 30 keV with a Quanta 600 SEM instrument using the STEM detector and analyzed with the ImageJ software.

The ^{29}Si and ^{13}C cross-polarization (CP)/magic angle spinning (MAS) NMR experiments were carried out on a Bruker Avance DSX NMR spectrometer with a 9.4 T magnet (79.51 MHz ^{29}Si Larmor frequency, 100.64 MHz ^{13}C Larmor frequency), while the ^{11}B NMR experiments were acquired on a Bruker Avance NMR spectrometer with a 16.45 T magnet (224.67 MHz ^{11}B Larmor frequency). For the ^{29}Si NMR experiments, samples were spun at 5.00 kHz in rotors of 7 mm diameter. Scans (1600 total) were accumulated with CP excitation that used a 2 ms CP pulse duration with ramped proton power and TPPM decoupling. The recycle delays of 2.24 s were chosen to be five times the proton spin lattice relaxation times. CP conditions were set with Kaolin, whose resonances also served as secondary chemical shift standard at −91.34 ppm (center between doublet). For the ^{13}C CP/MAS NMR experiments, samples were spun at 10.00 kHz in rotors of 4 mm diameter. Scans (880 total) were accumulated with CP excitation using a 1 ms CP pulse duration with ramped proton power and TPPM decoupling using recycle delays of 3 s. CP power conditions were set on glycine, where carbonyl resonances also served as secondary chemical shift standard at 176.06 ppm. For ^{11}B NMR experiments, samples were spun at 10 and 25 kHz to determine center bands and to identify spinning sidebands in rotors of 2.5 mm diameter. The $NaBH_4$ resonance served as a secondary chemical shift standard at −42.1 ppm relative to $BF_3 \cdot Et_2O$. For ^{11}B NMR spectra 1800 scans were accumulated, with a pulse length of 0.55 μs at pulse power of 40 kHz. Pulse repetition times of 1 s were used. Because of the substantial boron background the spectrum of an empty rotor acquired under the same conditions were subtracted for analysis.

High resolution powder diffraction patterns were collected on an Empyrean from PANalytical. X-rays were generated from a copper target with a scattering wavelength of 1.54 Å. The step-size of the experiment was 0.02°.

DSC was performed with a TA Instruments Q Series calorimeter. Weighed samples were hermetically sealed in aluminum pans and the temperature was scanned between −80 and +200 °C at a rate of 10 °C min^{-1}, a refrigerated cooling systems (RCS) served as coolant. Glass transition regions were then determined from the DSC traces as a midpoint temperature.

Conductivity was measured as follows. All samples were vacuum-dried at 50 °C overnight to minimize the effects of residual water absorption. Sample thicknesses were determined with a micrometer, and were in the range 80–750 μm. Gold contacts 3 mm or 4 mm in diameter were deposited directly onto the polymer films by means of a Polaron E5100 sputter coater. Sputtering was accomplished in 3 minutes per side. After gold electrodes were deposited, each sample was placed into a measurement cell and attached to an impedance analyzer. An oscillation level of 1 V was applied over the frequency range of 10 Hz to 10 MHz. For intermediate frequencies in this range, samples were modeled as a parallel circuit comprised of a resistor and capacitor. The impedance experiments were performed from 25 °C to 75 °C, then back to 25 °C, stabilizing the temperature at 5° intervals in each direction.

Transference measurements were performed as previously described[39] on a series of samples made by methods C and D. Samples were dried in a vacuum oven overnight and were then placed in an argon-filled glove box where they were subsequently mounted in a conductance cell. The conductance cell consisted of sandwiching the polymer between two non-blocking electrodes comprised of pieces of lithium metal ribbon (0.38 mm thick). Experiments measured the isothermal (25 °C) transient ionic current (ITIC) profiles during charging and charge-reversal cycles over the range of 1 to 2 V.

3 Results and discussion

3.1 Local structure: NMR and FTIR

To characterize the local structure of the SPEs and analyze the incorporation of boron, ^{29}Si (Fig. 1 and 3), ^{11}B (Fig. 2), and ^{13}C CP MAS NMR spectra were recorded. The ^{29}Si NMR spectrum of the reference sample with 0% TEB contains six signals: T^1 {C–SiO(O$^-$)$_2$}, T^2 {C–SiO$_2$(O$^-$)}, T^3 {C–SiO$_3$}, Q^2 {SiO$_2$(O$^-$)$_2$}, Q^3 {SiO$_3$(O$^-$)}, and Q^4 {SiO$_4$}, where the upper index indicates the number of Si–O–Si bridging oxygens of the three (T-species) or four (Q species) oxygens at the silicon nucleus. Upon incorporation of TEB, intensity of the T^3 species signal increases with increasing TEB content, while that of the T^2 species decreases, which indicates that boron increases the silica network formation due to formation of either Si–O–Si or Si–O–B bonds or both (Fig. 1). It is noteworthy that the chemical shifts of the signals do not change, thus we are unable to identify Si–O–B bonds.

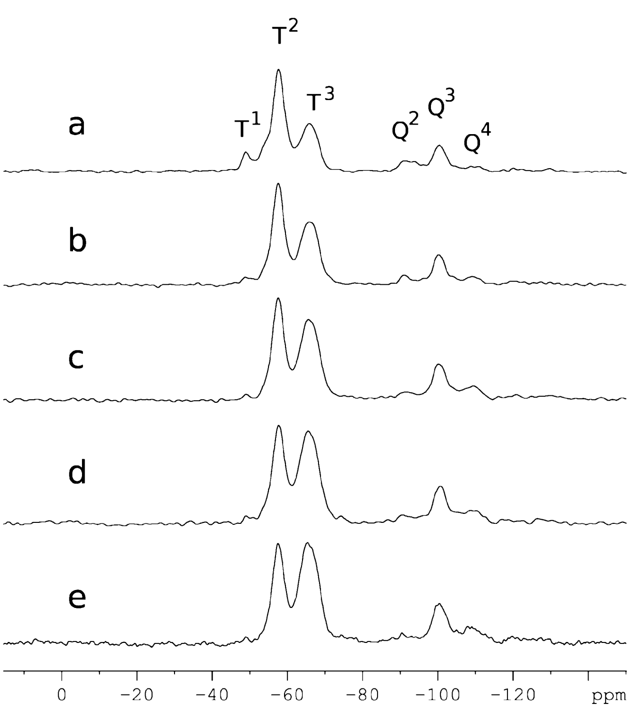


Fig. 1 ^{29}Si CP/MAS NMR spectra of SPEs with (a) 0% TEB, (b) 5% TEB, (c) 10% TEB, (d) 15% TEB, and (e) 20% TEB prepared by method C.

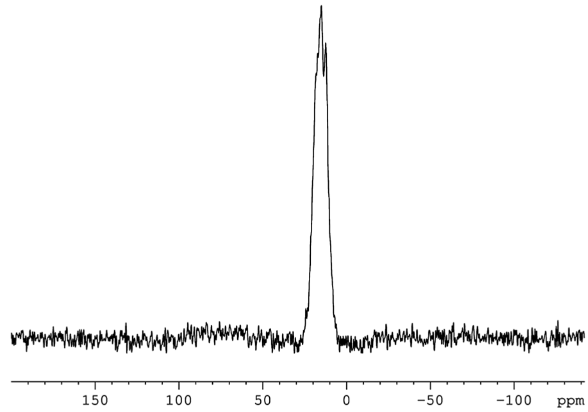


Fig. 2 ^{11}B MAS NMR spectrum of the SPE with 20% of TEB prepared according to method C.

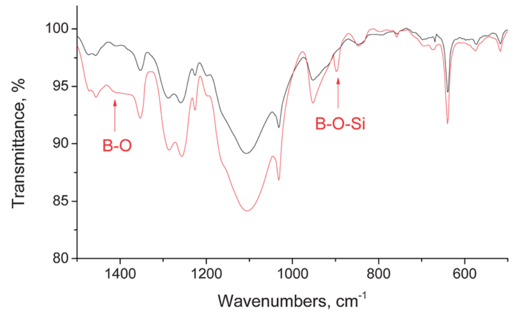


Fig. 3 FTIR spectra of the reference SPE sample with 0% TEB (black) and the SPE based on the OIC with 25% TEB (red).

Previous reports confirm the absence of the ^{29}Si chemical shift differences between Si–O–Si and Si–O–B bonds in organically modified silicates,[50] which thus precludes interpretation of Si–O–B bonds with ^{29}Si NMR. Assessment of such bonds was reported for similar materials with high boron content with ^{11}B MAS NMR.[51] However, because of the low weight percentage of boron in the SPEs reported here, even for the case with 20–30 mol% of TEB added to OIC, information obtained from ^{11}B NMR is limited. The ^{11}B NMR spectrum of the SPE with 20 mol % TEB (Fig. 2) shows that after incorporation into OIC the boron species retain tricoordinate structure, and thus Lewis acid character toward anions, but reliable data on the boron environment are not available.

To identify Si–O–B bonds, FTIR measurements were carried out. Comparison of FTIR spectra of the reference SPE sample (0 mol% TEB) and the SPE based on the OIC with 25 mol% TEB (Fig. 3) shows that the latter displays a shoulder at ~1400 cm^{-1} which can be associated with the formation of the B–O bonds and a band at 890 cm^{-1}. This vibration can be assigned to the B–O–Si bridges,[51] thus clearly confirming formation of the Si–O–B bonds.

Carbon NMR spectra of SPEs with various amounts of TEB and obtained by different methods are identical and show major resonances, assigned to the PEO groups (70.1 ppm),[52] –CH$_2$– groups of reacted GLYMO (26.1 and 23.2 ppm), and

–C–Si– units (9 ppm).[53] Absence of signals at 44 and 51 ppm characteristic of the glycidyl group of GLYMO suggests that this group fully reacts during SPE formation, which promotes crosslinking of OIC and PEG.[54]

Since the desired outcome is to place the majority of boron species on the exterior of the OIC particle to allow the strong interactions with triflate anions, different methods of TEB addition were explored as indicated in the Experimental section 2.2. In method A, TEB was added at low temperature after OIC was fully hydrolyzed and precondensed to allow attachment of boron species on the surface of the OIC particles. In method B, TEB was added in the mixture of all reagents before hydrolysis and condensation, thus location of boron species should be determined by reactivity of all substances. We assume that reactivity of TEB, TEOS and GLYMO are comparable, thus the boron species could be spread through the bulk of the OIC particle. Method C is similar to method B, but the hydrolysis was carried out at low temperature to slow down hydrolysis and condensation. This might result in more even distribution of species compared to method B. Finally, in method D, TEB was added into the OIC which was already prehydrolyzed with 15% HCl. Here boron species might attach to already partially formed OIC particles and thus would be unevenly spread, but probably closer to the OIC NP surface. The OIC particles prepared by all four methods are depicted in Scheme 2. It is worth noting, however, that this is a simplified representation of the OIC particles after precondensation in mild conditions (see Experimental). Final condensation is achieved by a thermal treatment at 70 °C overnight and 130 °C for one hour when the precondensed OIC is embedded in the polymer. These steps could cause both migration of the boron species and coalescence of the OIC particles, thus altering the model representations in Scheme 2.

The ^{29}Si CP MAS NMR spectra presented in Fig. 4 show that the highest degree of Si species condensation, *i.e.*, the highest intensity of the T^3 species, is achieved with method B when TEB is added at the same time with all reagents. Comparison of this spectrum with that of the reference SPE without boron species (Fig. 1a) clearly shows a noticeable increase of the T^3 signal intensity at the cost of the T^2 species, which reveals that boron species enhance condensation. For the method C sample, a slightly lower intensity of the T^3 signal is observed compared to that of method B, while for the A and D samples, the T^3 signal intensities are lower but still much higher than that of the reference sample (Fig. 1a). Thus, in all cases, the addition of TEB results in a higher degree of Si species condensation. Location of boron species can be indirectly assessed through the electrochemical performance of SPEs, which is discussed in Sections 3.3–3.5.

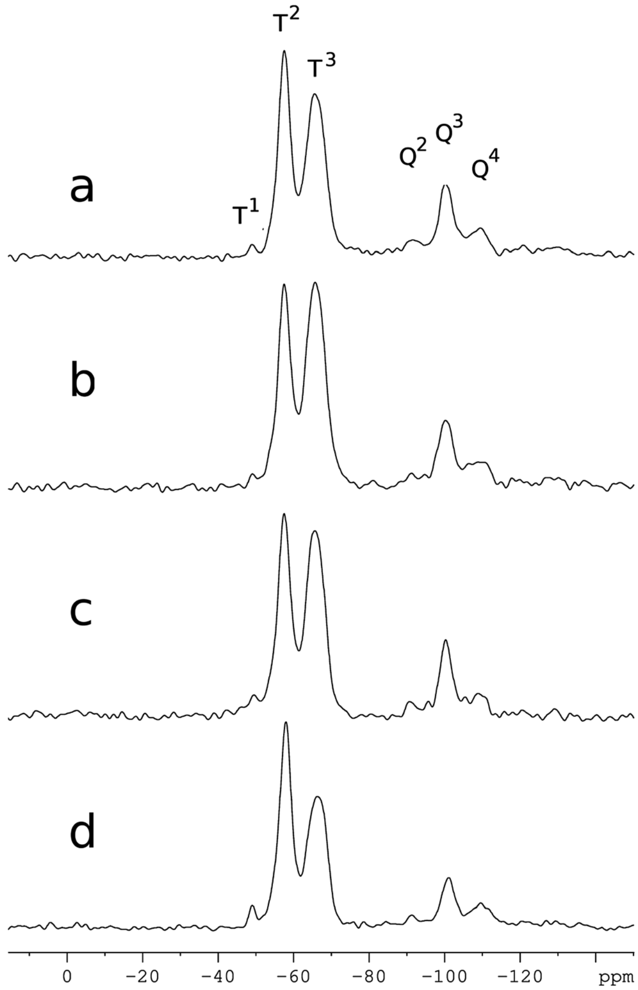
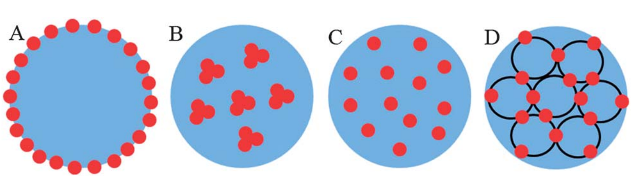


Fig. 4 ^{29}Si CP MAS NMR spectra of SPEs with 10% of TEB added according to methods A (a) B (b), C (c), and D (d).



Scheme 2 Schematic representations of OIC particles (blue) with boron species (red).

3.2 Mobility and crystallinity: DSC and XRD

To estimate mobility of the PEG polymer chains and their crystallization in the composite SPEs when boron species are incorporated, DSC measurements were performed (Table 1).

Fig. 5a shows that the DSC trace of the SPE with no boron species is practically featureless except for a glass transition. This is consistent with our earlier data on SPEs based on 600 Da PEG.[38,39] The SPEs which contain boron moieties in OIC show unusual DSC traces, the typical example of which is presented in Fig. 5b. A pronounced, broad endothermic transition is observed with a peak temperature at about 100 °C. Unlike this transition, melting of the PEG crystalline phase is normally represented by a narrow peak with a melting temperature of about 60 °C.[40,55] The high resolution XRD profile of the same sample presented in Fig. 6 shows a typical amorphous pattern with a strong peak at $2\theta = 22.5°$ featuring the approximate distance of 4.44 Å (according to the Bragg equation) for the

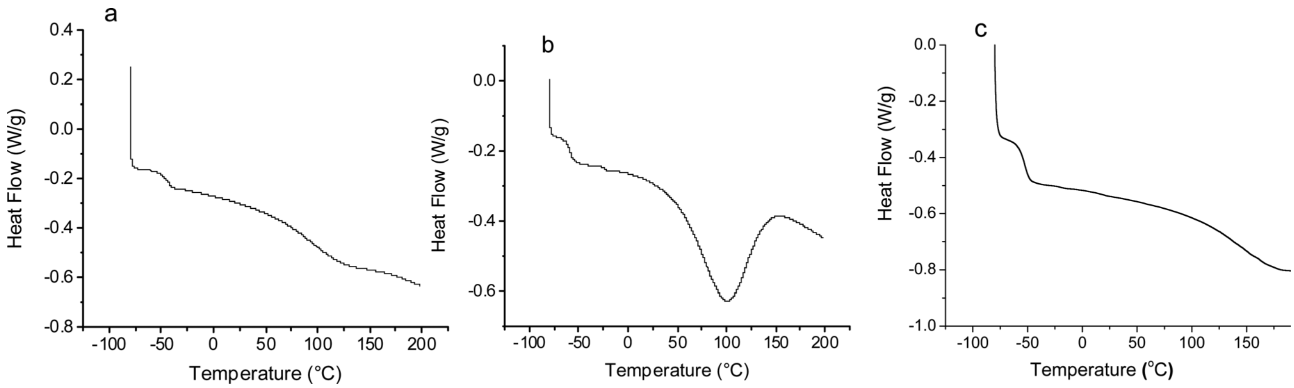


Fig. 5 DSC traces of SPEs which contain 55% OIC without boron species (a), with 10 mol% TEB added by method C (b) and the same sample after additional drying in a vacuum oven at room temperature overnight (c). For each sample, three scans were taken, but each time for a fresh portion of the material.

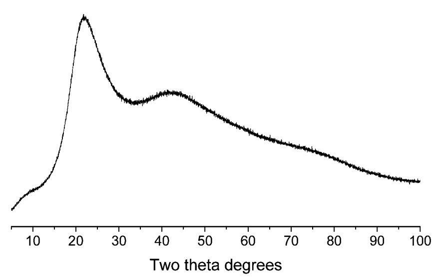


Fig. 6 XRD profile of SPE based on 55% of OIC with 10 mol% of TEB added by method C.

nearest neighbor and two following peaks which likely stem from second and third nearest neighbors.

Considering that the sample is amorphous, the endothermic transition in the DSC trace (Fig. 5b) could be assigned to water removal. Apparently tricoordinate boron species in OIC enhance water retention or adsorption, while SPEs without boron species do not show this trend. Overnight drying of the boron-containing SPE sample in vacuum at room temperature allows complete removal of water, evidenced by the observation that the DSC trace after drying exhibited only a glass transition (Fig. 5c). It is worth noting that for conductivity measurements, the samples were dried overnight in vacuum and stored in a dessicator before measurements.

The data in Table 1 show that glass transition temperatures (T_g) in all the samples vary between −47 and −60 °C. The lower the T_g, the higher the mobility of the polymer segments at ambient temperature. The observation that attachment of polymer chains to a solid surface results in a T_g increase due to a decrease of polymer chain mobility is well documented.[56] Here, 600 Da PEG is a liquid at room temperature, while the SPE films are solid rubber-like materials. Thus, similar to the SPEs described in previous papers[38,39] tethering of the PEG end groups to the OIC NP surface takes place *via* glycidyl group polymerization which should limit the PEO mobility. In the case of addition of TEB by method C, T_g first decreases and then increases (Table 1) when the TEB content increases. Thus, incorporation of low amounts of boron species (5–10 mol% TEB) leads to the diminished interaction of PEG chains with OIC particles followed by increased interactions related to increase in boron content. These changes might be caused by the change of the OIC particle sizes (see Section 3.3).

3.3 Conductivity and morphology of the OIC particles using STEM

Incorporation of boron species should increase the conductivity of SPEs.[42] Moreover, the higher the fraction of boron species at the interface with PEG-Li triflate, the higher the conductivity might be expected. However, the comparison of conductivities of the SPE samples prepared with 10 mol% TEB by four different methods (see Section 3.1) showed puzzling results: method A where we expected the highest fraction of the boron species in the OIC particle exterior, did not lead to any conductivity increase. A similar result was obtained for method B. On the other hand, in method C the conductivity increased by a factor of five, while in method D, conductivity increased by a factor of two. These data demonstrate that the method of the TEB addition does influence the conductivities of SPEs, but not in the way we expected. To understand this phenomenon, we evaluated the OIC particle size in all SPEs. As demonstrated earlier, OIC particle size significantly influences the electrochemical performance of composite SPEs.[39]

In this work for the first time we used STEM to determine sizes of the OIC particles in the native SPE environment, *i.e.*, without calcination at 500 °C in oxidizing media as described previously.[39] A caveat of this characterization is the brief exposure of SPE samples to water during sectioning for STEM imaging. This brief exposure could lead to PEG swelling and OIC particle aggregation, thus particle distribution within a SPE section cannot be determined. On the other hand, this moisture-induced effect does not alter the size of the OIC particles; therefore, particle sizes and morphologies can be assessed. Representative images obtained from the STEM measurements are shown in Fig. 7. Mean OIC particle sizes and particle size distributions of all samples are presented in Table 1.

The size analysis of the SPEs prepared by method C shows some correlation between the OIC particle size and conductivity. The conductivity showed practically no change upon

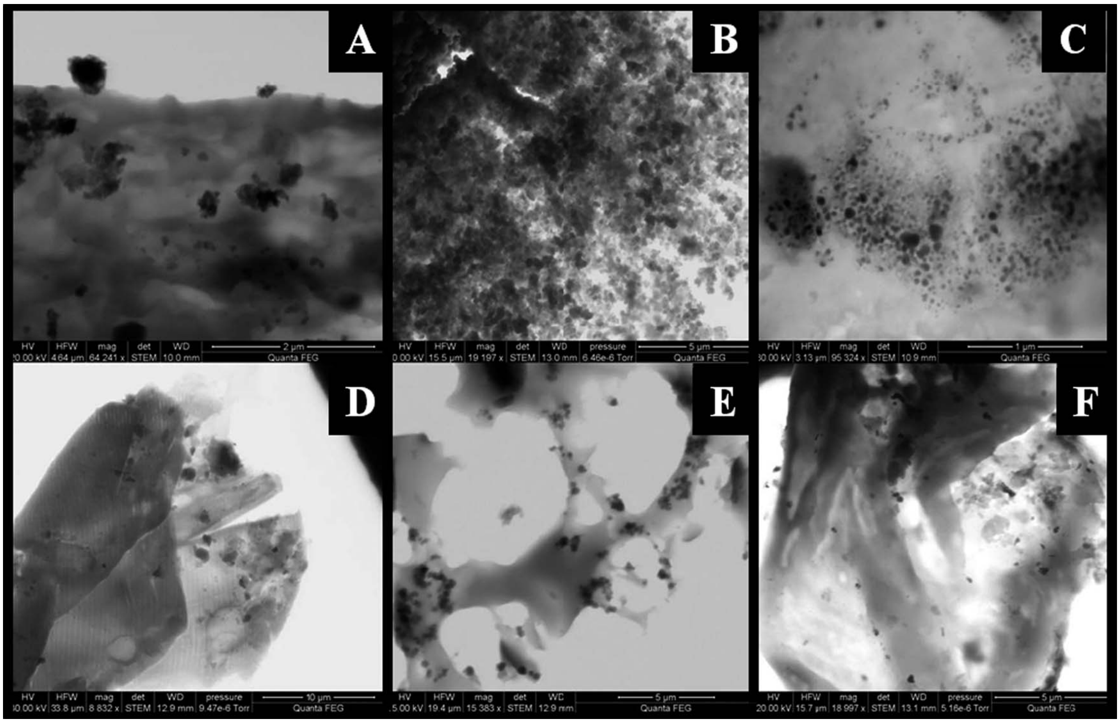


Fig. 7 STEM images of the SPE samples with different amounts of TEB added to OIC by method C: without TEB (A), 5 mol% TEB (B), 10 mol% TEB (C), 15 mol% TEB (D), 20 mol% TEB (E), and 25 mol% TEB (F).

addition of 5 mol% TEB and the OIC particle size did not change either. Addition of 10 mol% TEB led to a drastic decrease of the OIC particle size and a dramatic increase of conductivity. Further increase of the TEB amount to 15 mol% resulted in larger particles and lower conductivity. In the range of 20–30 mol% TEB even larger particles were formed although no conductivity change was observed. Apparently, only boron species located at the interface between OIC particles and polymer-in-salt component influence the conductivity. Thus, despite the increase in total amount of boron content, the amount of boron moieties which are exposed to interaction with anions apparently decreases, which leads to lower conductivity when the TEB amount increases beyond 10 mol%.

Comparison of the SPEs prepared by four different methods at 10 mol% of TEB showed a similar trend. For methods A and B, the OIC particle size is similar to that of the reference sample without boron species and conductivities values were similar as well. For methods C and D, the OIC particles obtained were smaller and the conductivities were higher. Therefore, the method of TEB addition does influence the electrochemical properties of the SPEs but mainly through control of the OIC particle size. Most likely, the addition of 10 mol% TEB by method C makes nanoparticles resistant to coalescence, thus affords an optimal condition to produce particles with small size and large surface area.

3.4 Temperature dependence of conductivity

Thermal dependence of conductivity was measured for several SPE samples of greatest interest, which included a reference sample (0 mol% TEB) and SPE samples prepared from methods C and D with 10 mol% TEB (Fig. 8). Conductivities measured follow an approximately Arrhenius temperature dependence, proportional to $\exp[-E_0/RT]$ with activation energy E_0 of 46.5, 51.9, and 58.3 kJ mol^{-1} for SPEs prepared by methods C, D, and reference sample, respectively.

3.5 Transference measurements

During electrochemical cycling, the ITIC experiments exhibited two time scales, a short time scale during which the current peaked (assigned to the lithium ions, τ_+) and a longer time scale during which the current decreased (assigned to the anions, τ_-). The lithium ion transference number, T_+, was determined from the ratio of the corresponding ionic mobilities (with assumption of equal densities of anions and cations) and the values obtained are presented in Table 2. The value for the reference sample is consistent with previous results.[39]

For moderate amounts of added TEB, lithium transference increases significantly as compared to the reference sample. The peak transference at 10 mol% TEB does not differ between the samples prepared by either method C or D. Doubling the amount of TEB resulted in a decrease in transference to values similar to the reference sample. Comparisons of conductivities and lithium transference numbers (Tables 1 and 2) show high conductivities and high Li transference numbers are not necessarily correlated. For example, for the sample prepared by method C with 5 mol% TEB, the conductivity is as low as that of the reference sample. At the same time, the Li transference number of the former is nearly equal to that of the sample

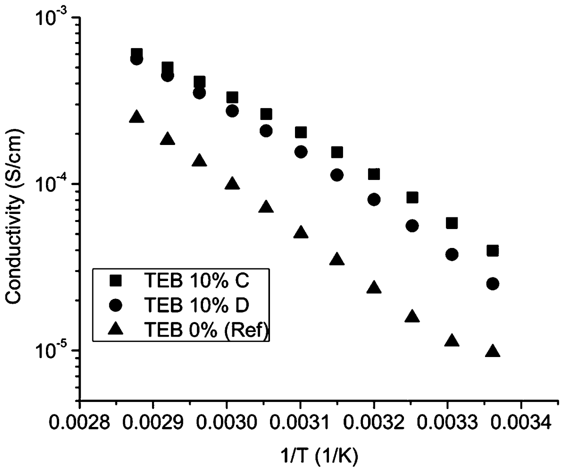


Fig. 8 Arrhenius plot of the temperature dependence of the conductivity.

Table 2 Lithium transference numbers of the SPEs studied

Method	TEB content, mol%	T_+ ($\pm$0.05)
—	0	0.76
C	5	0.88
C	10	0.89
C	20	0.79
D	10	0.89

prepared by method C with 10 mol% TEB and displaying the highest conductivity. We believe the lithium ion transference increases for 5 mol% TEB primarily due to decreased mobility of the anions rather than from increased mobility of the lithium ions. In the case of the samples with 10 mol% TEB prepared by methods C and D, transference numbers measured are the same, while conductivities differed by a factor of 2.5. We believe that here again the decreased mobility of anions is the same, which leads to high transference numbers and reflects a similar amount of boron species in the OIC NP exterior. However, larger OIC particles in the sample prepared by method D impede Li ion mobility. A significant decrease of Li transference number for the sample with 20 mol% TEB compared to that with 10 mol % TEB reflects boron species buried within the OIC bulk, thus the anion mobility is less hindered.

Conclusions

We suggested a novel approach to incorporate boron species to SPEs by reaction of a strong Lewis acid, TEB, with OIC NPs which are proposed to have stronger interactions with triflate anions than silica sites, to result in enhanced lithium ion conductivity and transference numbers. We demonstrated that both the concentration of boron species and the method of the TEB addition influence the morphology of the OIC obtained and the subsequent electrochemical behavior of SPEs. Comparisons of conductivity and lithium ion transference number measurements demonstrate that enhanced transference is determined by decreased anion mobility, while for enhanced conductivity, facilitated Li ion mobility is also required. We demonstrated that the addition of 10 mol% of TEB in the reaction mixture when OIC NPs are formed allows formation of the smallest NPs, the highest conductivity and the lithium transference number of 0.89 which is unprecedented for non-single-ion conductors.

Notes and references

1 L. Ye and Z. Feng, in *Polymer Electrolytes*, ed. C. Sequeira and D. Santos, Woodhead Publishing Ltd., Cambridge, UK, 2010, pp. 550–582.
2 J. W. Fergus, *J. Power Sources*, 2010, **195**, 4554–4569.
3 B. Scrosati and J. Garche, *J. Power Sources*, 2010, **195**, 2419–2430.
4 F. Croce, G. B. Appetecchi, L. Persi and B. Scrosati, *Nature*, 1998, **394**, 456–458.
5 D. E. Fenton, J. M. Parker and P. V. Wright, *Polymer*, 1973, **14**, 589–590.
6 A. Ghosh and P. Kofinas, *J. Electrochem. Soc.*, 2008, **155**, A428.
7 R. L. Karlinsey, L. M. Bronstein and J. W. Zwanziger, *J. Phys. Chem. B*, 2004, **108**, 918–928.
8 K. Ichikawa, L. C. Dickinson, W. J. MacKnight, M. Watanabe and N. Otaga, *Polymer*, 1992, **33**, 4699.
9 Z. Zhang and S. Fang, *J. Appl. Polym. Sci.*, 2000, **77**, 2957.
10 H. R. Allcock, S. J. M. O'Connor, D. L. Olmeijer, M. E. Napierala and C. G. Cameron, *Macromolecules*, 1996, **29**, 7544.
11 Z. Florjanczyk, W. Kraweic, W. Wieczorek and M. Siekierski, *J. Polym. Sci., Polym. Phys.*, 1995, **33**, 629.
12 S. Rajendran, V. S. Bama and M. R. Prabhu, *Ionics*, 2010, **16**, 283–287.
13 J. L. Acosta and E. Morales, *J. Appl. Polym. Sci.*, 1996, **60**, 1185.
14 N. S. Choi, Y. G. Lee, J. K. Park and J. M. Younko, *Electrochim. Acta*, 2001, **46**, 1581–1586.
15 X. Hou and K. S. Siow, *Polymer*, 2000, **41**, 8689.
16 F. Latif, M. Aziz, N. Katun, A. M. M. Ali and M. Z. Yahya, *J. Power Sources*, 2006, **159**, 1401.
17 S. Ramesh, T. Winie and A. Arof, *Eur. Polym. J.*, 2007, **43**, 1963.
18 H. J. Rhoo, H. T. Kim, J. K. Park and T. S. Huang, *Electrochim. Acta*, 1997, **42**, 1571–1579.
19 F. Croce, R. Curini, A. Martinelli, L. persi, F. Ronchi and B. Scrosati, *J. Phys. Chem. B*, 1999, **103**, 10632.
20 B. Scrosati, F. Croce and L. Persi, *J. Electrochem. Soc.*, 2000, **147**, 1718–1721.
21 Y. Karatas, R. D. Banhatti, N. Kaskhedikar, M. Burjanadze, K. Funke and H.-D. Wiemhofer, *J. Phys. Chem. B*, 2009, **113**, 15473–15484.
22 V. Aravindan and P. Vickraman, *Polym. Eng. Sci.*, 2009, **49**, 2109–2115.
23 F. Kaneko, S. Wada, M. Nakayama, M. Wakihara, J. Koki and S. Kuroki, *Adv. Funct. Mater.*, 2009, **19**, 918–925.
24 H. Aydin and A. Bozkurt, *J. Appl. Polym. Sci.*, 2012, **124**, 1193–1199.

25 C. G. Joo, L. M. Bronstein, R. L. Karlinsey and J. W. Zwanziger, *Solid State Nucl. Magn. Reson.*, 2002, **22**, 235–246.
26 B. K. Mandal, C. J. Walsh, T. Sooksimuang, S. J. Behroozi, S.-G. Kim, Y.-T. Kim, E. S. Smotkin, R. Filler and C. Castro, *Chem. Mater.*, 2000, **12**, 6–8.
27 H. Tsutsumi, M. Yamamoto, M. Morita, Y. Matsuda, T. Nakamura and H. Asai, *J. Power Sources*, 1993, **41**, 291–298.
28 S. Bayoudh, N. Parizel and L. Reibel, *Polym. Int.*, 2000, **49**, 703–711.
29 L. C. Hardy and D. F. Shriver, *J. Am. Chem. Soc.*, 1985, **107**, 3823–3828.
30 M. Watanabe, Y. Suzuki and A. Nishimoto, *Electrochim. Acta*, 2000, **45**, 1187–1192.
31 R. Meziane, J.-P. Bonnet, M. Courty, K. Djellab and M. Armand, *Electrochim. Acta*, 2011, **57**, 14–19.
32 Y. S. Zhu, X. W. Gao, X. J. Wang, Y. Y. Hou, L. L. Liu and Y. P. Wu, *Electrochem. Commun.*, 2012, **22**, 29–32.
33 E. P. Giannelis, *Adv. Mater.*, 1996, **8**, 29–35.
34 J. P. Lemmon, J. Wu, C. Oriakhi and M. M. Lerner, *Electrochim. Acta*, 1995, **40**, 2245–2249.
35 M. Kurian, M. E. Galvin, P. E. Trapa, D. R. Sadoway and A. M. Mayes, *Electrochim. Acta*, 2005, **50**, 2125–2134.
36 J. C. Hutchison, R. Bissessur and D. F. Shriver, *Chem. Mater.*, 1996, **8**, 1597–1599.
37 L. M. Bronstein, R. L. Karlinsey, B. Stein, Z. Yi, J. Carini and J. W. Zwanziger, *Chem. Mater.*, 2006, **18**, 708–715.
38 L. M. Bronstein, C. Joo, R. L. Karlinsey and J. W. Zwanziger, *Chem. Mater.*, 2001, **13**, 3678–3684.
39 L. M. Bronstein, R. L. Karlinsey, B. Stein and J. W. Zwanziger, *Solid State Ionics*, 2005, **176**, 559–570.
40 L. M. Bronstein, R. L. Karlinsey, K. Ritter, C. G. Joo, B. Stein and J. W. Zwaziger, *J. Mater. Chem.*, 2004, **14**, 1812–1820.
41 L. M. Bronstein, R. L. Karlinsey, B. Stein, Z. Yi, J. Carini and J. W. Zwanziger, *Chem. Mater.*, 2006, **18**, 708–715.
42 F. Kaneko, S. Wada, M. Nakayama, M. Wakihara and S. Kuroki, *ChemPhysChem*, 2009, **10**, 1911–1915.
43 G. Zukowska, M. Szczechura, M. Marcinek, A. Zubrowska, A. Adamczyk-Wozniak, A. Sporzynski and W. Wieczorek, *ECS Trans.*, 2009, **16**, 105–114.
44 T. Itoh, M. Yoshikawa, T. Uno and M. Kubo, *Ionics*, 2009, **15**, 27–33.
45 M. A. Mehta and T. Fujinami, *Chem. Lett.*, 1997, 915–916.
46 X. Sun and C. A. Angell, *Electrochim. Acta*, 2001, **46**, 1467–1473.
47 N. Matsumi, K. Sugai and H. Ohno, *Macromolecules*, 2002, **35**, 5731–5733.
48 N. Matsumi, K. Sugai and H. Ohno, *Macromolecules*, 2003, **36**, 2321–2326.
49 Y. Nishihara, M. Miyazaki, Y. Tomita, Y. Kadono and K. Takagi, *J. Polym. Sci., Polym. Chem.*, 2008, **46**, 7913–7918.
50 P. Innocenzi, G. Brusatin and F. Babonneau, *Chem. Mater.*, 2000, **12**, 3726–3732.
51 G. D. Soraru, N. Dallabona, C. Gervais and F. Babonneau, *Chem. Mater.*, 1999, **11**, 910–919.
52 J. Ma, C. Guo, Y. Tang, J. Wang, L. Zheng, X. Liang, S. Chen and H. Liu, *J. Colloid Interface Sci.*, 2006, **299**, 953–961.
53 L. M. Bronstein, C. Joo, R. Karlinsey, A. Ryder and J. W. Zwanziger, *Chem. Mater.*, 2001, **13**, 3678–3684.
54 M. Templin, U. Wiesner and H. W. Spies, *Adv. Mater.*, 1997, **9**, 814–817.
55 X.-Y. Yu, M. Xiao, S.-J. Wang, Q.-Q. Zhao and Y.-Z. Meng, *J. Appl. Polym. Sci.*, 2010, **115**, 2718–2722.
56 B. P. Grady, A. Paul, J. E. Peters and W. T. Ford, *Macromolecules*, 2009, **42**, 6152–6158.

